

Lab 1: Inspecting Table Size

Query Query History

```
1 SELECT
2     relname AS object_name,
3     pg_size_pretty(pg_table_size(fed.oid)) AS data_size,
4     pg_size_pretty(pg_indexes_size(fed.oid)) AS index_size,
5     pg_size_pretty(pg_total_relation_size(fed.oid)) AS total_size
6 FROM pg_class fed
7 JOIN pg_namespace n
8     ON n.oid = fed.relnamespace
9 WHERE fed.relkind = 'r'
10    AND n.nspname = 'public'
11 ORDER BY pg_total_relation_size(fed.oid) DESC;
12
```

Data Output Messages Notifications

Showing rows: 1 to 15 Page No: 1 of 1

	object_name name	data_size text	index_size text	total_size text
1	rental	1232 kB	1120 kB	2352 kB
2	payment	896 kB	920 kB	1816 kB
3	film	736 kB	200 kB	936 kB
4	film_actor	272 kB	216 kB	488 kB
5	inventory	232 kB	208 kB	440 kB
6	customer	96 kB	112 kB	208 kB
7	address	88 kB	64 kB	152 kB
8	city	64 kB	48 kB	112 kB
9	film_category	72 kB	40 kB	112 kB
10	actor	40 kB	32 kB	72 kB
11	store	8192 bytes	32 kB	40 kB
12	staff	16 kB	16 kB	32 kB
13	language	8192 bytes	16 kB	24 kB
14	category	8192 bytes	16 kB	24 kB
15	country	8192 bytes	16 kB	24 kB

Total rows: 15 Query complete 00:00:00.080 LF Ln 12, Col 1

Lab 2: The Impact of a New Index

Step 1 : Check current size of 'payment' indexes

13 SELECT pg_size_pretty(pg_indexes_size('payment'));

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	pg_size_pretty text
1	920 kB

Total rows: 1 Query complete 00:00:00.058 LF Ln 13, Col 51

Step 2 : Create a heavy index on payment_date

```
14 CREATE INDEX idx_payment_date_long ON payment ( payment_date , amount ,customer_id );
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 39 msec.

Total rows: Query complete 00:00:00.039 LF Ln 14, Col 86

Step 3 : Check size again

```
16 SELECT pg_size_pretty ( pg_indexes_size ( 'payment ' ) ) ;
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	pg_size_pretty
1	1488 kB

Total rows: 1 Query complete 00:00:00.042 LF Ln 16, Col 58

How many Pages (8KB) did this new index consume?

The new index consumed approximately $1488 - 920 = 568$ KB, which corresponds to $568 / 8 = 71$ pages (8 KB per page)

Lab 3: Partial Index Practice

Create Partial Index

```
18 CREATE INDEX idx_expensive_films
19 ON film (replacement_cost)
20 WHERE replacement_cost > 25.00;
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 48 msec.

Total rows: Query complete 00:00:00.048 LF Ln 20, Col 32

Find the size of this specific index

```
22 SELECT
23     indexrelname AS index_name,
24     pg_size_pretty(pg_relation_size(indexrelid)) AS index_size
25 FROM pg_stat_user_indexes
26 WHERE indexrelname = 'idx_expensive_films';
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	index_name	index_size
1	idx_expensive_fil...	16 kB

Total rows: 1 Query complete 00:00:00.045 LF Ln 26, Col 44

```
32 CREATE INDEX idx_full_replacement_cost
33 ON film (replacement_cost);
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 44 msec.

Total rows: Query complete 00:00:00.044 LF Ln 33, Col 28

```
35 SELECT
36     indexrelname AS index_name,
37     pg_size_pretty(pg_relation_size(indexrelid)) AS index_size
38 FROM pg_stat_user_indexes
39 WHERE indexrelname = 'idx_full_replacement_cost';
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	index_name name	index_size text
1	idx_full_replacement_c...	40 kB

Total rows: 1 Query complete 00:00:00.055 LF Ln 39, Col 50

Partial index: 16 kB

Full index: 40 kB

Storage saved = 40 - 16 = 24 kB

-> The partial index is 16 KB, while the full index is 40 KB. Therefore, the partial index saves 24 KB of storage, representing a 60% (24/40 x 100%) reduction in index size.